

Homework 4

Merit and Excellence problem solving · about 60 minutes

Each question graded N0–E8

Show all working, and finish every proof with a sentence. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

- 1 Find the discriminant of $x^2 - 4x + 4 = 0$. How many roots are there? [A]
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- 2 Solve $\log_2 x = 5$. [A]
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- 3 Factorise $6x^2 + x - 2$. [A]
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- 4 Write a quadratic with roots $\frac{1}{2}$ and 3, with whole-number coefficients. [A]
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QUESTION ONE

- (a) Show that the sum of four consecutive whole numbers is always even. [A]
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- (b) Prove that the product of an odd number and an even number is always even. [A]
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- (c) Prove that $(n + 3)^2 - (n - 3)^2$ is always a multiple of 12. [M]
.....
- (d) Sam was asked to prove that the square of any odd number is one more than a multiple of 8. **Sam's answer has 2 errors.** Find each one, fix it, and say why it loses marks. [M]

Sam's answer

$3^2 = 9 = 8 + 1$ and $5^2 = 25 = 24 + 1$, so it is true.
Also $(2n + 1)^2 = 4n^2 + 1$.

- (e) On a calendar (7 days a row), a 3-by-3 square has top-left number n . Prove that the sum of all nine numbers is always 9 times the middle number. [E]

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QUESTION TWO

- (a) Find the points where $y = x^2 - 4$ meets $y = 3x$. [A]

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- (b) Show that the line $y = 2x - 5$ never meets the curve $y = x^2 + 1$. [M]

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- (c) Find the values of m for which $y = mx + 1$ is a tangent to $y = x^2 + 2x + 5$. [M]

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- (d) Find every point where $x^2 + y^2 = 20$ meets $y = x^2$. [E]

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QUESTION THREE

- (a) Solve $\sqrt{2x + 1} = 5$. [A]

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- (b) Solve $\sqrt{x + 12} = x$. [M]

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A rectangular pen is built against a wall, so 30 m of fencing makes only three sides: two of width x m and one of length $(30 - 2x)$ m.

(c) The pen has area 100 m^2 . Find its possible dimensions. **[M]**

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(d) Show that the pen cannot have area 120 m^2 . **[M]**

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(e) Use the discriminant to find the largest area the pen can enclose, and its dimensions. **[E]**

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